

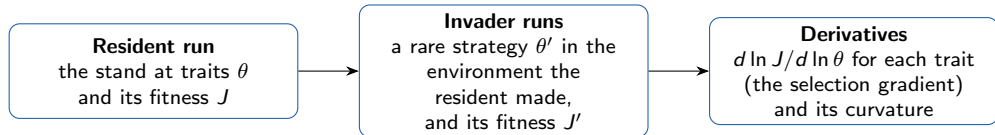
## Fitness gradients for TF24 in plant

What changed in plant, odelia and phylloptim, what it costs and what it buys

8 October 2026

## An evolutionary analysis needs fitness and its derivatives, many times over

regnans finds and classifies singular strategies from plant runs of three kinds.



- One singular strategy in one trait takes about **84 such runs**. A resident run on a 40-year record takes about 90 s.
- Until now plant returned  $J$  alone, so every derivative came from repeated runs.

Run counts: docs/design-grid-controller.md, "Where the arc starts".

## The whole analysis step, end to end, in plant's R interface

```
p    <- add_strategies(scm_base_parameters("TF24"), trait_matrix(0.32, "lma"))
p$node_schedule_times <- list(seq(0, 39.6, length.out = 108)) # 108 introductions over 40 years

base <- control_tf24(3e-5, Control(node_density_in_birth_date = TRUE,      # parts 1, 5, 6, 8
                                   ode_split_sign_changes = TRUE))        # part 4
pilot <- run_scm(p54, env, control_tf24(1e-3), record_trajectory = TRUE) # p at 54 introductions
ctrl  <- control_window(pilot, list(lma_half, lma_double), base = base)  # part 7

scm <- run_scm(p, env, ctrl, record_trajectory = TRUE)
g    <- stand_gradient(scm, "offspring_production") # all 49 derivatives in one pass, part 2
scm$run_mutant(invader)                          # an invader on the resident's recording, part 3
d    <- diagnose_scm(p, env, ctrl, invaders)        # error estimate and failures, part 9
```

On the 40-year long-drought record the middle block takes **209 s**: 37 s for the pilot, 40 s for the resident run and 132 s for its gradient. It returns  $J$  and all 49 elasticities. The same answers by brute force take 886 s. Every function above is on the branches. The next slide lists the nine parts, and the deck takes them in order.

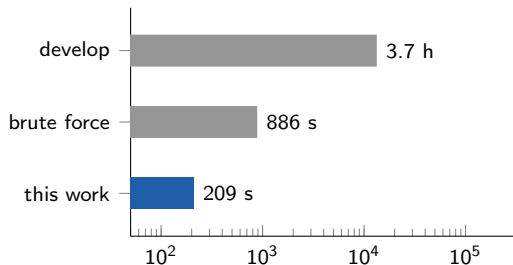
Timed alone, one run each: docs/measurements/headline/.

## What was built: nine parts

|   | what was built                                                        | the problem it fixes                             | in plant                            |
|---|-----------------------------------------------------------------------|--------------------------------------------------|-------------------------------------|
| 1 | Establishment integrated exactly between cohorts                      | $J$ depended on when cohorts were introduced     | the birth-date coordinate           |
| 2 | Every derivative from one reverse pass                                | 98 runs per gradient by differences              | <code>stand_gradient()</code>       |
| 3 | An invader replays the resident's recording exactly                   | events skipped; no gradient for an invader       | <code>run_mutant()</code>           |
| 4 | A cohort's step cut where its net production changes sign             | gradients shaped like staircases                 | <code>ode_split_sign_changes</code> |
| 5 | Tolerance factors, and TF24's preset                                  | wasted steps, refused gradients, failed invaders | <code>control_tf24()</code>         |
| 6 | The soil stepped on its own                                           | the soil's stability set 93% of the steps        | <code>ode_soil_alone_share</code>   |
| 7 | A tolerance factor that rises as fitness is earned, read from a pilot | late steps spent accuracy on nothing             | <code>control_window()</code>       |
| 8 | Each interval's leaf area on eight interpolated crowns                | a run could not estimate its own error           | the birth-date light field          |
| 9 | Each run reports its error and failures                               | no error estimate; one failure ended an analysis | <code>diagnose_scm()</code>         |

Part 1 makes the birth-date coordinate's integrals exact. Parts 2 and 3 make gradients possible, and part 4 makes them smooth in the traits. Parts 5 to 7 make each run cheaper, and parts 8 and 9 let a run say how accurate it is. Each part closes one issue on the aornugent forks and is off by default.

## Altogether: the same answers in a quarter to a half of brute force's time



seconds for  $J$  and its 49 derivatives, long drought

**develop** plant's develop with its default Control1. One run takes 134 s; its gradient needs 100 runs by differences, so this bar is computed.

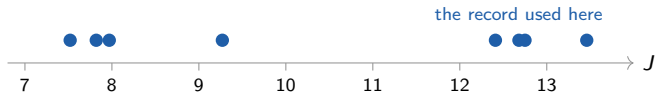
**brute force** this work's code with parts 4 to 7 off: a tight tolerance ( $1e-5$ ) and twice the introductions (215).

**this work** the code on slide 3, pilot included.

- On the episodic record the ratio is also 0.24 (128 s against 536 s), and all 50 quantities lie within  $0.40\varepsilon$  of brute force's. The next two slides define  $\varepsilon$ .
- On long drought 108 introductions are too few, and 215 match brute force's accuracy at 0.43 of its time. Near the end, a waterfall shows what each part contributes.

## How close is close enough? The weather sets the scale

Eight 40-year rainfall records were drawn from one long-drought climate. They differ only in the day-to-day weather. The same stand, run on each, earns these fitnesses  $J$ :



- Nothing about the stand changed between these runs, yet  $J$  ranges from 7.5 to 13.5. In logs, their standard deviation is **0.25**.
- A numerical error far below that spread cannot change what an analysis concludes about this climate. One of comparable size could.
- So the tolerance for  $\ln J$  is set at a tenth of the spread:  $\varepsilon = 0.025$ , or 2.5% of  $J$ .

## Each derivative gets its own tolerance by the same rule

The same eight records give each elasticity a spread too. Its  $\varepsilon$  is a tenth of its standard deviation, and never below 0.01.

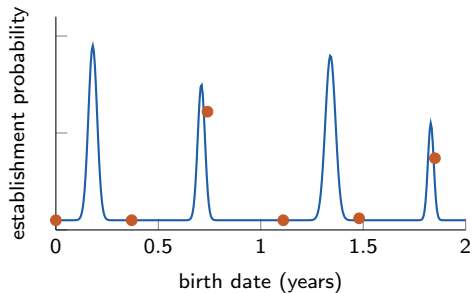
| quantity, for the resident              | mean over 8 records | standard deviation | $\varepsilon$ |
|-----------------------------------------|---------------------|--------------------|---------------|
| $\ln J$                                 | 2.32                | 0.25               | 0.025         |
| elasticity of $J$ in <code>lma</code>   | -9.6                | 0.87               | 0.087         |
| elasticity of $J$ in <code>a_dG2</code> | 1.10                | 0.19               | 0.019         |

- The rest of the deck reports errors as multiples of  $\varepsilon$ . “Within  $0.4\varepsilon$ ” means within 0.4 of each quantity’s own tolerance.
- *Accurate* means within  $\varepsilon$  of the converged answer.
- *Reproducible* means a 5% change of tolerance, or moving every introduction by a quarter of its spacing, moves each quantity by less than  $\varepsilon/3$ .
- Both are checked on five records: constant rain, long-wet, episodic, dry and long drought.

## Part 1 of 9: establishment integrated exactly between cohorts

|   | what was built                                                        | the problem it fixes                             | in plant                            |
|---|-----------------------------------------------------------------------|--------------------------------------------------|-------------------------------------|
| 1 | Establishment integrated exactly between cohorts                      | $J$ depended on when cohorts were introduced     | the birth-date coordinate           |
| 2 | Every derivative from one reverse pass                                | 98 runs per gradient by differences              | <code>stand_gradient()</code>       |
| 3 | An invader replays the resident's recording exactly                   | events skipped; no gradient for an invader       | <code>run_mutant()</code>           |
| 4 | A cohort's step cut where its net production changes sign             | gradients shaped like staircases                 | <code>ode_split_sign_changes</code> |
| 5 | Tolerance factors, and TF24's preset                                  | wasted steps, refused gradients, failed invaders | <code>control_tf24()</code>         |
| 6 | The soil stepped on its own                                           | the soil's stability set 93% of the steps        | <code>ode_soil_alone_share</code>   |
| 7 | A tolerance factor that rises as fitness is earned, read from a pilot | late steps spent accuracy on nothing             | <code>control_window()</code>       |
| 8 | Each interval's leaf area on eight interpolated crowns                | a run could not estimate its own error           | the birth-date light field          |
| 9 | Each run reports its error and failures                               | no error estimate; one failure ended an analysis | <code>diagnose_scm()</code>         |

## The problem: establishment changes within days, cohorts are months apart



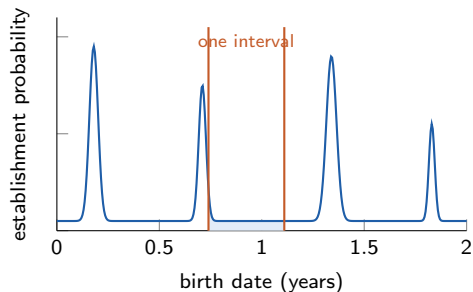
Shape for illustration.

plant#94 (PLANT-93).

On the birth-date coordinate each cohort stands for the seedlings born between it and the next cohort. Its density was the birth rate times the establishment probability at its own birth date (the orange points), and integrals over birth date were trapezia through those samples.

TF24's establishment rises after rain and falls as the soil dries, within days. Cohorts are introduced weeks to months apart, so the samples catch some pulses and miss others.  $J$  then depended on *when* cohorts were introduced, as well as on how many there were.

## How it works: each cohort carries the establishment of its interval as a state



Shape for illustration.

1. While a cohort is the newest, it integrates the establishment probability over its interval as an ODE state, together with that integral's first moment. The solver's own steps resolve each pulse.
2. When the next cohort is introduced, the interval closes and its integrals stop changing.
3. An integral over birth date weights each cohort by establishment integrated against its hat function. Only the rest of the integrand, which is smooth, is interpolated between cohorts.

A cohort's density is then no longer a state of its own.

## The result: establishment is exact, whatever the introductions

- The establishment term is integrated to the solver's tolerance, however the cohorts fall relative to the rain. What remains of the node error comes from the smooth part of the integrand.
- Before, a run's error estimate from its nodes' values saw only 4–5% of the true error: seedlings born into a closed establishment window made steps between nodes that no nodal value shows. Parts 8 and 9 rely on that cause being gone.
- The height coordinate is unchanged, and K93, which establishes every seed, gives the same answers to ten digits.
- On this coordinate `Species$log_densities_state` is removed and `refine_schedule()` raises an error.

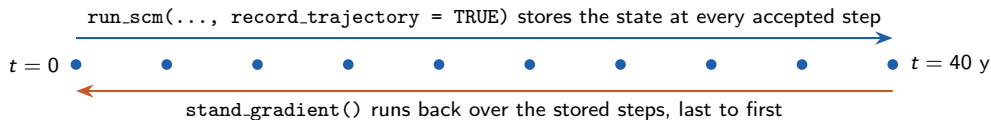
## Part 2 of 9: every derivative from one reverse pass

|   | what was built                                                        | the problem it fixes                             | in plant                            |
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| 2 | Every derivative from one reverse pass                                | 98 runs per gradient by differences              | <code>stand_gradient()</code>       |
| 3 | An invader replays the resident's recording exactly                   | events skipped; no gradient for an invader       | <code>run_mutant()</code>           |
| 4 | A cohort's step cut where its net production changes sign             | gradients shaped like staircases                 | <code>ode_split_sign_changes</code> |
| 5 | Tolerance factors, and TF24's preset                                  | wasted steps, refused gradients, failed invaders | <code>control_tf24()</code>         |
| 6 | The soil stepped on its own                                           | the soil's stability set 93% of the steps        | <code>ode_soil_alone_share</code>   |
| 7 | A tolerance factor that rises as fitness is earned, read from a pilot | late steps spent accuracy on nothing             | <code>control_window()</code>       |
| 8 | Each interval's leaf area on eight interpolated crowns                | a run could not estimate its own error           | the birth-date light field          |
| 9 | Each run reports its error and failures                               | no error estimate; one failure ended an analysis | <code>diagnose_scm()</code>         |

## The problem: a gradient by differences costs two runs per parameter

- A central difference runs the stand at  $\theta_j(1 \pm h)$  for each parameter  $j$ . TF24 has 49 differentiable parameters, so one gradient costs about **98 runs**.
- An analysis needs a gradient for the resident and one for each invader, at every step of its search.
- Where moving a trait moves an event, such as a cohort reaching maturity between two steps, the difference does not converge as  $h$  shrinks.

## How it works: the run records its steps, and one pass runs back over them



At each stored step, from the last to the first, the pass:

1. re-evaluates the step's six Runge–Kutta stages, with XAD recording the arithmetic;
2. takes  $\lambda$ , the derivative of  $\ln J$  in the state at the step's end, and turns it into the derivative in the state at the step's start;
3. adds that step's contribution to  $\partial \ln J / \partial \theta$  for all 49 parameters at once.

Where a cohort was introduced, the pass carries  $\lambda$  into the newborn's initial size, so seed traits get their share. The leaf's inner solve is never recorded: `phyloptim` supplies its derivatives directly. Only doubles cross into R.

## The result: all 49 derivatives for the cost of 2.5 runs

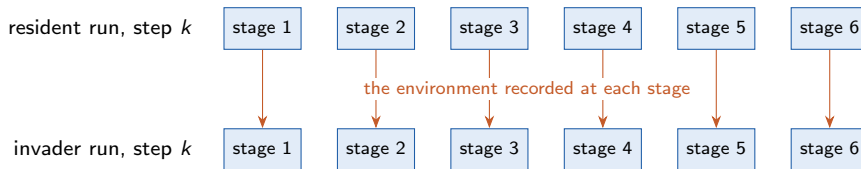
- One pass costs **2.5–2.7 runs**, whatever the number of introductions, against about 98 runs by differences.
- It agrees with central differences to  $7e-4$  of the resident's `lma` elasticity, and closer for the other entries checked.
- A derivative that does not exist is refused with a stated reason; a NaN without one would be a bug.

`docs/measurements/perf-adjoint.md`; `eps-spread.md`.

## Part 3 of 9: an invader replays the resident's recording exactly

|   | what was built                                                        | the problem it fixes                             | in plant                            |
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| 5 | Tolerance factors, and TF24's preset                                  | wasted steps, refused gradients, failed invaders | <code>control_tf24()</code>         |
| 6 | The soil stepped on its own                                           | the soil's stability set 93% of the steps        | <code>ode_soil_alone_share</code>   |
| 7 | A tolerance factor that rises as fitness is earned, read from a pilot | late steps spent accuracy on nothing             | <code>control_window()</code>       |
| 8 | Each interval's leaf area on eight interpolated crowns                | a run could not estimate its own error           | the birth-date light field          |
| 9 | Each run reports its error and failures                               | no error estimate; one failure ended an analysis | <code>diagnose_scm()</code>         |

## What plant already did: an invader runs on the resident's recording



- An invader is rare, so it does not change the light or water the resident made.
- `scm$run_mutant(invader)` on develop already pins the invader to the resident's step times and reads the environment the resident recorded at every Runge–Kutta stage.
- So an invader at the resident's own traits should return the resident's fitness exactly. The next slide shows where it did not, and what stopped its gradient being taken.

SCM::run\_mutant on develop: `r_set_ode_times(patch.step_history)`.

## The problem: the replay was not exact, and could not be differentiated

- **Events were skipped.** The replay applied each entry's introductions but not its rain pulses, harvests or climate extremes, and after an entry it read the environment from before the entry's events.
- **The recorded light was rebuilt from heights**, which can land one unit in the last place off. TF24's leaf solve amplified that to  $5 \times 10^{-9}$  in a log density.
- **An invader kept no recording**, so its own reverse pass had nothing to run back over. `store_trajectory()` re-ran the invaders as residents.
- On the gradient branch, an invader was also evaluated at the resident's leaf operating points, and two invaders together failed.

| an invader at the resident's traits: its $\ln J'$ less the run's $\ln J$ | before                 | now |
|--------------------------------------------------------------------------|------------------------|-----|
| no events                                                                | $-7.1 \times 10^{-15}$ | 0   |
| a rain pulse of 0.05 at $t = 2.5$                                        | $-1.2 \times 10^{-7}$  | 0   |
| a 50% harvest at $t = 3.5$                                               | +0.69                  | 0   |
| a lethal climate extreme at $t = 3.5$                                    | +1.3                   | 0   |

plant:#95, on test-mutant.R's TF24 stand.

## How it works: one map per schedule entry, and every evaluation repeated

- **One map for each schedule entry**, `Patch::apply_insertion(t)`: the entry's events in schedule order, then its introductions. The run, the invader's replay and the reverse pass all apply this one map, so they cannot disagree about an entry.
- **The light is recorded as the interpolant holds it**: its knot values, slopes and canopy top. A replay restores those numbers instead of rebuilding them.
- **Every evaluation at a recorded state repeats the one recorded**. The invader reads the recorded environment and solves its own leaf.
- **The invader stores its own steps**, so `stand_gradient()` runs back over them with the resident's environment held fixed, which is what a selection gradient asks for.

PLANT-95 (plant#95), odelia a05f5c2.

## The result: an exact replay, with its own gradient

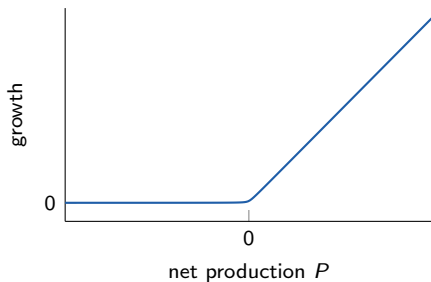
- An invader at the resident's own traits returns  $J' = J$  to the last bit, with or without events. The tests check this on every record.
- Two invaders together each have exactly the fitness they have alone.
- An invader's reverse pass agrees with differences of its own fitness to  $10^{-9}$  in `hmat` and `k_I`.
- The first invader reuses the resident's recording instead of re-running it: 139 s against 300 s (PLANT-99).
- An invader run costs about 0.77 of a resident run.

plant#95, #99.

## Part 4 of 9: a cohort's step cut where its net production changes sign

|   | what was built                                                        | the problem it fixes                             | in plant                            |
|---|-----------------------------------------------------------------------|--------------------------------------------------|-------------------------------------|
| 1 | Establishment integrated exactly between cohorts                      | $J$ depended on when cohorts were introduced     | the birth-date coordinate           |
| 2 | Every derivative from one reverse pass                                | 98 runs per gradient by differences              | <code>stand_gradient()</code>       |
| 3 | An invader replays the resident's recording exactly                   | events skipped; no gradient for an invader       | <code>run_mutant()</code>           |
| 4 | A cohort's step cut where its net production changes sign             | gradients shaped like staircases                 | <code>ode_split_sign_changes</code> |
| 5 | Tolerance factors, and TF24's preset                                  | wasted steps, refused gradients, failed invaders | <code>control_tf24()</code>         |
| 6 | The soil stepped on its own                                           | the soil's stability set 93% of the steps        | <code>ode_soil_alone_share</code>   |
| 7 | A tolerance factor that rises as fitness is earned, read from a pilot | late steps spent accuracy on nothing             | <code>control_window()</code>       |
| 8 | Each interval's leaf area on eight interpolated crowns                | a run could not estimate its own error           | the birth-date light field          |
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## The problem: growth has a corner where net production crosses zero

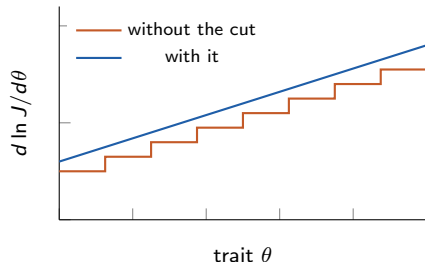


TF24 grows a cohort by the positive part of its net production  $P$ . Below zero a cohort draws on storage and does not grow.

The positive part is smoothed over a band of  $10^{-4}$  kg/yr. A cohort crosses that band in a small fraction of one step: in the median crossing the step is  $5 \times 10^4$  times longer than the bend.

So at the scale of a step, the cohort's rate has a corner.

## The problem: a corner inside a step turns the gradient into a staircase



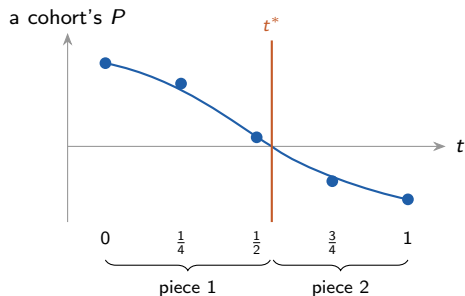
Shape for illustration.

A Runge–Kutta step evaluates the rate at six stages inside the step. Its error from a corner depends on where the corner sits among those stages.

As  $\theta$  moves, each crossing slides through the step. Each time it passes a stage the error jumps, so the gradient moves in steps. A 5% change of tolerance moves every step boundary, and so moves the gradient by up to  $0.52\varepsilon$ .

A curvature taken across one of those jumps was off by  $13\varepsilon$ .

## How it works: find the crossing inside the step, and integrate that cohort in two pieces



After odelia accepts a step, it hands plant the step just taken, with an interpolant of the whole state across it. With `ode_split_sign_changes = TRUE`, plant then:

1. reads each cohort's  $P$  at five fractions of the step,  $0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$  and  $1$ , and finds where the curve through them crosses zero;
2. for each cohort that crosses, integrates that cohort alone from the step's start to  $t^*$  and from  $t^*$  to the end, in the light and water read off the interpolant;
3. leaves every other cohort and the soil on the whole step.

## How it works in reverse: the cut moves with the traits

- In the reverse pass, each piece is a step of its own: its stages are re-evaluated and recorded like any other step's.
- The cut time  $t^*$  is where  $P(t^*; \theta) = 0$ , so it moves as  $\theta$  moves. The pass differentiates it through that equation (the implicit function theorem), so  $dt^*/d\theta$  enters the gradient.
- plant owns which cohort is cut, the field it reads and its pieces. One body of code serves the forward run and the reverse pass, so the two cannot disagree about a cut. odelia owns the arithmetic: the interpolant, the pieces and the root-finding.

## The result: smooth, reproducible gradients for 5–9% more time

- $\ln J$  lies 3–13 $\times$  nearer the converged answer wherever cohorts cross zero.
- A 5% change of tolerance moves no gradient entry by more than  $\varepsilon/6$ , against up to  $0.52\varepsilon$  before.
- The reverse pass through the cuts agrees with central differences to  $2e-3$  on every record.
- The leaf's solve now stops at roundoff (phylloptim#17), so re-running a step at a trait  $10^{-12}$  away moves  $\ln J$  smoothly. Its noise fell from  $2.7e-8$  to about  $1e-14$ .
- The cuts cost 5–9% of a gradient run.

## Part 5 of 9: tolerance factors, and TF24's preset

|   | what was built                                                        | the problem it fixes                             | in plant                            |
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| 5 | Tolerance factors, and TF24's preset                                  | wasted steps, refused gradients, failed invaders | <code>control_tf24()</code>         |
| 6 | The soil stepped on its own                                           | the soil's stability set 93% of the steps        | <code>ode_soil_alone_share</code>   |
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| 9 | Each run reports its error and failures                               | no error estimate; one failure ended an analysis | <code>diagnose_scm()</code>         |

## How plant accepts a step, and what a tolerance factor does

plant's integrator accepts a step when, for every state  $y_i$ , its estimated error is below its allowed error:

$$|\text{error}_i| \leq f_i(t) (\text{tol\_rel} |y_i| + \text{tol\_abs})$$

- $f_i(t)$  is the state's **tolerance factor**, new in this work and 1 by default. A factor of 10 lets that state carry ten times the error, so it stops forcing short steps. Nothing else changes: the same equations, the same stepper.
- A factor can be fixed for a state, or vary with patch age  $t$ . This work uses both:
  - the soil layers' factor, fixed at 10 (next slide);
  - a factor on every state that rises late in the patch's life, read from a pilot run (part 7);
  - a cap of 100 on their product, for every state (the slide after next).

`odelia OdeControl::adjust_step_size`. In `Control`: `ode_weight_soil`, `ode_weight_times` and `_factors`, `ode_weight_max`; to be renamed for tolerance factors.

## The soil's tolerance factor: 10

- On long drought the soil's stability limit forces 16 447 of the run's 17 684 steps, yet the soil carries little of  $J$ 's error.
- With the soil layers' factor at 10, a gradient run takes 21% less time, and both the resident's and the invader's derivatives still pass the accuracy and reproducibility tests at tolerance  $3e-5$ .
- 20 is the largest factor that still converges. At 30 and 50 the saturated top layer cycles at the explicit stability limit.
- Part 6 goes further, taking the soil out of the step entirely where the plants barely draw on it.

docs/grid-dynamics.md §10; docs/design-grid-controller.md, phase 1a.

## The cap: no state's tolerance factor above 100

The factors multiply. With part 7's factor  $F(t)$ , which rises to 100 late in the patch, each state's tolerance factor is

$$f_{\text{soil}}(t) = \min(10 F(t), 100), \quad f_{\text{other}}(t) = \min(F(t), 100).$$

- Without the cap the soil's factor reached 1000. An accepted step could then carry a soil layer to its water-potential ceiling, where TF24's gradient is undefined, and every gradient on every record was refused.
- With the cap every gradient is finite, and the soil's own error test stays in force.

`state-weights (ode.weight_max); docs/grid-dynamics.md §12.`

## control\_tf24() sets TF24's values once

| setting                                            | why                                                                                                                                                                                                                                                |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\text{tol\_abs} = 10^{-4} \times \text{tol\_rel}$ | Tied this way, $J$ 's error stays within about half of $\text{tol}$ . With plant's default absolute tolerance it was about $20\times$ larger at $\text{tol} = 1\text{e-}4$ .                                                                       |
| the soil's tolerance factor 10                     | This part.                                                                                                                                                                                                                                         |
| tolerance factors capped at 100                    | This part.                                                                                                                                                                                                                                         |
| steps no longer than 15 days                       | An invader's storage pools go unstable on steps longer than 26 days. On the episodic and dry records, the invader at twice the resident's $\text{lma}$ failed on 27-day steps. With the limit none fails, for at most 2.4% of the window's saving. |
| the soil stepped alone below a 10% share           | Part 6.                                                                                                                                                                                                                                            |

It does not set the window's factor, which depends on the analysis: part 7 adds it to this preset from a pilot run. plant's own `Control()` defaults are unchanged.

PLANT-104, PLANT-105 (plant#104, #105).

## Part 6 of 9: the soil stepped on its own

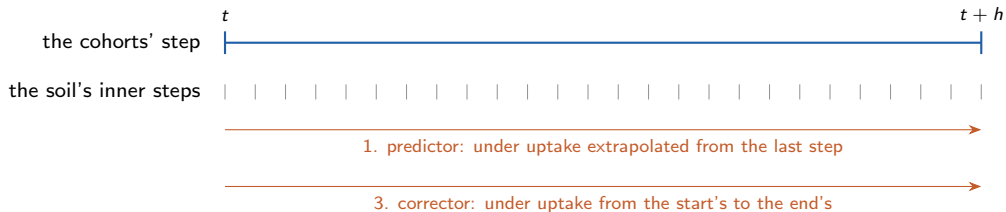
|   | what was built                                                        | the problem it fixes                             | in plant                            |
|---|-----------------------------------------------------------------------|--------------------------------------------------|-------------------------------------|
| 1 | Establishment integrated exactly between cohorts                      | $J$ depended on when cohorts were introduced     | the birth-date coordinate           |
| 2 | Every derivative from one reverse pass                                | 98 runs per gradient by differences              | <code>stand_gradient()</code>       |
| 3 | An invader replays the resident's recording exactly                   | events skipped; no gradient for an invader       | <code>run_mutant()</code>           |
| 4 | A cohort's step cut where its net production changes sign             | gradients shaped like staircases                 | <code>ode_split_sign_changes</code> |
| 5 | Tolerance factors, and TF24's preset                                  | wasted steps, refused gradients, failed invaders | <code>control_tf24()</code>         |
| 6 | The soil stepped on its own                                           | the soil's stability set 93% of the steps        | <code>ode_soil_alone_share</code>   |
| 7 | A tolerance factor that rises as fitness is earned, read from a pilot | late steps spent accuracy on nothing             | <code>control_window()</code>       |
| 8 | Each interval's leaf area on eight interpolated crowns                | a run could not estimate its own error           | the birth-date light field          |
| 9 | Each run reports its error and failures                               | no error estimate; one failure ended an analysis | <code>diagnose_scm()</code>         |

## The problem: the soil's stability sets almost every step

- Water moves between TF24's soil layers within days. An explicit Runge–Kutta step longer than that makes the soil unstable, so the step must stay short.
- On long drought the soil alone forces **16 447 of the run's 17 684 steps**. The cohorts on their own would need about 9200.
- For much of a run the stand draws only a small share of the water moving through the soil, so the soil hardly depends on the plants over a single step.

docs/grid-dynamics.md §1; docs/design-grid-controller.md, "The time axis".

## How it works: the soil takes short inner steps inside each step



At a step's start plant computes the uptake's share of the water moving through the soil. If it is under `ode_soil_alone_share` (0.1 in the preset), odelia:

1. integrates the soil alone across the step in short inner steps, under the plants' uptake extrapolated from the last step;
2. takes the cohorts' Cash–Karp step, each stage reading the soil from step 1;
3. integrates the soil again, under the uptake the step found, and keeps that end;
4. counts the gap between the two soil passes in the step's error.

The run records the extrapolation and the inner steps, so a replay and the reverse pass take the same steps.

## The result: a gradient run takes 0.63 of the time

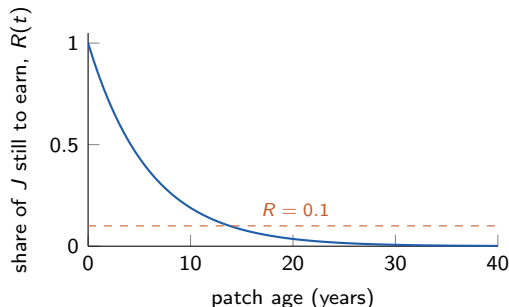
- Timed alone, a gradient run takes 376 s against 596 s.
- Steps fall 42–45% on the pulsed records and 64% under constant rain.
- Every elasticity moves by less than  $0.063\epsilon$ , and  $\ln J$  stays within  $5.8\text{e-}5$  of a reference at tolerance  $1\text{e-}6$  on all five records.
- It replaces an implicit solver for the soil (ARK) that was built first: under constant rain it costs 0.97 of that solver's steps, and the ARK branches are retired.

docs/measurements/soil-alone/prereg.txt, gates G1–G7.

## Part 7 of 9: a tolerance factor that rises as fitness is earned

|   | what was built                                                        | the problem it fixes                             | in plant                            |
|---|-----------------------------------------------------------------------|--------------------------------------------------|-------------------------------------|
| 1 | Establishment integrated exactly between cohorts                      | $J$ depended on when cohorts were introduced     | the birth-date coordinate           |
| 2 | Every derivative from one reverse pass                                | 98 runs per gradient by differences              | <code>stand_gradient()</code>       |
| 3 | An invader replays the resident's recording exactly                   | events skipped; no gradient for an invader       | <code>run_mutant()</code>           |
| 4 | A cohort's step cut where its net production changes sign             | gradients shaped like staircases                 | <code>ode_split_sign_changes</code> |
| 5 | Tolerance factors, and TF24's preset                                  | wasted steps, refused gradients, failed invaders | <code>control_tf24()</code>         |
| 6 | The soil stepped on its own                                           | the soil's stability set 93% of the steps        | <code>ode_soil_alone_share</code>   |
| 7 | A tolerance factor that rises as fitness is earned, read from a pilot | late steps spent accuracy on nothing             | <code>control_window()</code>       |
| 8 | Each interval's leaf area on eight interpolated crowns                | a run could not estimate its own error           | the birth-date light field          |
| 9 | Each run reports its error and failures                               | no error estimate; one failure ended an analysis | <code>diagnose_scm()</code>         |

The idea: late in a patch's life there is little fitness left to get wrong

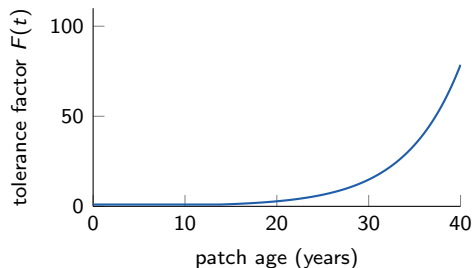


Shape for illustration.

Offspring are produced mostly early in a patch's life. Write  $R(t)$  for the share of the run's  $J$  still to be earned after time  $t$ .

A step late in the run can change  $J$  only through what is still to be earned. Holding it to the same tolerance as an early step buys accuracy  $J$  does not need.

The factor: every state's tolerance loosened by  $0.1/R(t)$



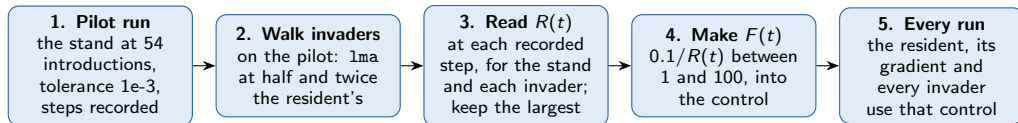
Shape for illustration.

$F(t) = 0.1/R(t)$ , kept between 1 and 100.

- $F$  stays 1 until only a tenth of  $J$  is left to earn, then rises as the rest is earned.
- It multiplies every state's tolerance factor from part 5, under the same cap of 100.
- A step takes the factor at its start time. In Control the factor is a table:  
`ode_weight_times` and `ode_weight_factors`.

$R(t)$  has to be known before the run that uses it.  
The next slide shows where it comes from.

Where  $R(t)$  comes from: a cheap pilot run of the same analysis



- Steps 2–4 are `control_window(pilot, invaders, base)`. It runs no new stand: it walks the invaders on the pilot and returns `base` with  $F$  added.
- The invaders matter because one that matures later earns later. Keeping the largest  $R$  means no step is loosened while any of them still has fitness to earn.
- At 54 introductions and  $1e-3$ , the pilot reads  $R$  within 10% wherever  $R \geq 10^{-3}$ . On long drought it takes 37–39 s, against 175 s for the run and gradient it serves.
- One pilot serves every run of the analysis around one resident. Under constant rain no pilot is taken:  $J$  is earned evenly, and the factor saves only 2%.

## How an analysis builds its control, parts 5 to 7 together

```
base <- control_tf24(3e-5, Control(node_density_in_birth_date = TRUE,
                                   ode_split_sign_changes = TRUE))
  # tied tolerance; the soil's factor 10; factors capped at 100;
  # steps no longer than 15 days; the soil alone below a 10% share

pilot <- run_scm(p54, env, control_tf24(1e-3), record_trajectory = TRUE) # step 1
ctrl  <- control_window(pilot, list(lma_half, lma_double), base = base) # steps 2-4

scm <- run_scm(p, env, ctrl, record_trajectory = TRUE) # step 5
```

Each step of every run then has, for each state,

$$\text{allowed error}_i = \min(f_i^{\text{soil}} F(t), 100) (\text{tol\_rel } |y_i| + \text{tol\_abs}),$$

with  $f_i^{\text{soil}} = 10$  for the soil layers and 1 for every other state, and steps no longer than 15 days. Where the plants draw under a tenth of the soil's water, the soil takes its own inner steps (part 6).

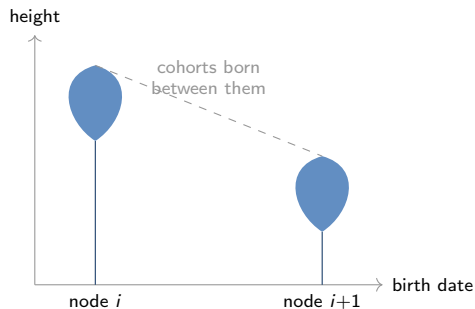
## The result: about a fifth fewer steps on the pulsed records

- The window's factor saves 21–23% of a run's steps on the pulsed records, and 2% under constant rain.
- On all five records nothing fails, and  $\ln J$  lies within  $5.8e-5$  of a reference at tolerance  $1e-6$ .
- Parts 5 and 7 together save 31–34% of a gradient run against the tied tolerance alone.

## Part 8 of 9: each interval's leaf area on eight interpolated crowns

|   | what was built                                                        | the problem it fixes                             | in plant                            |
|---|-----------------------------------------------------------------------|--------------------------------------------------|-------------------------------------|
| 1 | Establishment integrated exactly between cohorts                      | $J$ depended on when cohorts were introduced     | the birth-date coordinate           |
| 2 | Every derivative from one reverse pass                                | 98 runs per gradient by differences              | <code>stand_gradient()</code>       |
| 3 | An invader replays the resident's recording exactly                   | events skipped; no gradient for an invader       | <code>run_mutant()</code>           |
| 4 | A cohort's step cut where its net production changes sign             | gradients shaped like staircases                 | <code>ode_split_sign_changes</code> |
| 5 | Tolerance factors, and TF24's preset                                  | wasted steps, refused gradients, failed invaders | <code>control_tf24()</code>         |
| 6 | The soil stepped on its own                                           | the soil's stability set 93% of the steps        | <code>ode_soil_alone_share</code>   |
| 7 | A tolerance factor that rises as fitness is earned, read from a pilot | late steps spent accuracy on nothing             | <code>control_window()</code>       |
| 8 | Each interval's leaf area on eight interpolated crowns                | a run could not estimate its own error           | the birth-date light field          |
| 9 | Each run reports its error and failures                               | no error estimate; one failure ended an analysis | <code>diagnose_scm()</code>         |

## The problem: each interval's leaf area sat on its two end nodes' crowns

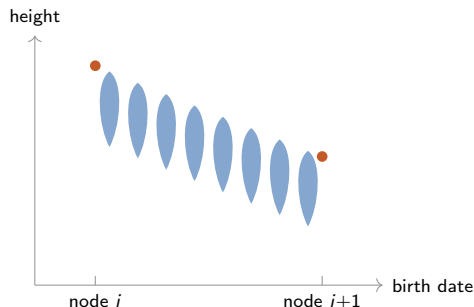


On the birth-date coordinate the light field sums each node's crown: its leaf area, spread below its top by the crown profile  $Q$ . Each node also stands for the cohorts born in the intervals either side of it.

So the leaf area of the cohorts born between two nodes was carried by the two nodes' own crowns, with their tops at the two nodes' heights. Those cohorts' own tops lie between the two (the dashed line), so the field below each node's top was off by an amount that depended on where the nodes happened to fall.

Its error did not fall at a fixed rate as nodes were added: the ratio of successive changes was 8.5 on long-wet and 6.6 on long drought, and 15–18 quantities changed sign. A run at half the introductions reported only 0.33–0.41 of the true error.

## How it works: each interval adds the crowns of eight interpolated cohorts



Each interval between two neighbouring nodes now contributes eight crowns, as if eight cohorts were born at evenly spaced dates across it. Each has the crown profile  $Q$ , with:

- its top at the height interpolated linearly between the two nodes' heights;
- its leaf area interpolated linearly between the two nodes' shares of the interval.

The crowns are read off the two nodes each time the light field is built, so no state is added. The newest interval, up to the boundary node, is formed once the boundary node is known. Where two cohorts cross in height, the crowns are summed tallest first, which keeps the light field's fast path (PLANT-107).

PLANT-107, PLANT-108 (plant#107, #108); Species::for\_each\_interval\_crown.

The result: the error falls fourfold per doubling, so it can be estimated

- With the interpolated crowns, the error falls with the square of the spacing. Measured ratios of successive changes are 3.5–4.0, against 4 in theory.
- That makes the error computable from two runs. With  $Q$  a quantity from a run and  $Q_{1/2}$  the same quantity at half the introductions,

$$\text{converged answer} \approx Q + \frac{Q - Q_{1/2}}{3}.$$

- The half-introduction run now reports 0.87–0.99 of the true error, against 0.33–0.41 before.
- The interpolated crowns cost 6.7% of a gradient run.
- An invader can be walked on some of the run's introductions, matched by birth date (PLANT-109).

## Part 9 of 9: each run reports its error and failures

|   | what was built                                                        | the problem it fixes                             | in plant                            |
|---|-----------------------------------------------------------------------|--------------------------------------------------|-------------------------------------|
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| 4 | A cohort's step cut where its net production changes sign             | gradients shaped like staircases                 | <code>ode_split_sign_changes</code> |
| 5 | Tolerance factors, and TF24's preset                                  | wasted steps, refused gradients, failed invaders | <code>control_tf24()</code>         |
| 6 | The soil stepped on its own                                           | the soil's stability set 93% of the steps        | <code>ode_soil_alone_share</code>   |
| 7 | A tolerance factor that rises as fitness is earned, read from a pilot | late steps spent accuracy on nothing             | <code>control_window()</code>       |
| 8 | Each interval's leaf area on eight interpolated crowns                | a run could not estimate its own error           | the birth-date light field          |
| 9 | Each run reports its error and failures                               | no error estimate; one failure ended an analysis | <code>diagnose_scm()</code>         |

## How it works: the run, repeated at every other and every fourth introduction



`diagnose_scm(p, env, ctrl, invaders)` runs the stand three times, keeping both ends, and walks each invader on each run. For every quantity ( $\ln J$ ,  $\ln J'$  and each elasticity) it returns:

- **error** =  $(Q - Q_{1/2})/3$ , the correction to the converged answer;
- **ratio** =  $(Q_{1/2} - Q_{1/4})/(Q - Q_{1/2})$ , near 4 where the estimate holds;

and, beside them, each invader's distance  $\ln(\theta'/\theta)$  in each parameter it moves, and each run, invader or gradient that failed, with its message.

## An example: three runs of a small test stand

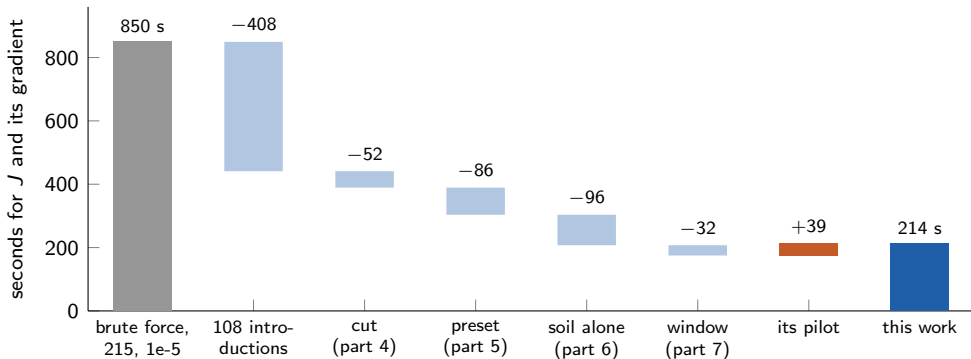
An eight-year stand at 9, 5 and 3 introductions, with one invader at 1.5 times the resident's `hmat` and one that cannot be built:

| run      | quantity          | value   | half    | quarter | error   | ratio |
|----------|-------------------|---------|---------|---------|---------|-------|
| resident | <code>ln J</code> | 4.7033  | 4.7979  | 5.1101  | -0.0315 | 3.30  |
| resident | <code>lma</code>  | -1.6028 | -1.5512 | -1.4264 | -0.0172 | 2.42  |
| late     | <code>ln J</code> | 5.5647  | 5.6787  | 6.0344  | -0.0380 | 3.12  |

| run    | stage | message                                                             |
|--------|-------|---------------------------------------------------------------------|
| broken | walk  | <code>L.tip</code> must be shorter than the birth-size flow path... |

- The ratios of 2.4–3.3 say nine introductions are too few to trust the estimate.
- The invader `late` is 0.405 away from the resident in `hmat`.
- The broken invader becomes three rows of failures, and the rest still answer. The two extra runs cost about three quarters of a run.

## Altogether: how the parts compose, on long drought



- Each bar adds one part to everything left of it, timed alone. The last configuration is the headline's: its  $J$  is the same to the bit.
- Parts 4 to 7 save 266 s and leave every quantity's error where it was: at 108 introductions the median elasticity stays within  $0.50\text{--}0.51\epsilon$  of the converged answer at each step.
- Halving the introductions saves the most and carries all of the remaining error. Parts 8 and 9 are what let a run see that error and decide whether it can afford the halving.

## Altogether: on long drought, 215 introductions match brute force at 0.43 of its time

The reference is extrapolated from this work's runs at 108 and 215 introductions,  $Q_{215} + (Q_{215} - Q_{108})/3$ . Each row compares  $\ln J$  and the 49 elasticities with it, in units of each quantity's  $\varepsilon$ . Time is relative to brute force's.

| long drought                         | $\ln J$           | elasticities, median | largest           | time |
|--------------------------------------|-------------------|----------------------|-------------------|------|
| brute force: 215 introductions, 1e-5 | $0.05\varepsilon$ | $0.12\varepsilon$    | $0.49\varepsilon$ | 1    |
| this work, 215 introductions         | $0.05\varepsilon$ | $0.13\varepsilon$    | $0.53\varepsilon$ | 0.43 |
| this work, 108 introductions         | $0.20\varepsilon$ | $0.50\varepsilon$    | $2.1\varepsilon$  | 0.24 |

- At 108 introductions the remaining error comes from the nodes. The half-introduction run reports it, so a run can tell that it needs more introductions.
- On the episodic record 108 introductions suffice: every quantity lies within  $0.40\varepsilon$  of brute force's.
- No run, invader or gradient fails on any of the five records.

## What changes for code that uses plant today

- Every new behaviour sits behind a `Control` field or a new function, and is off by default. With it off, results are bit-identical, apart from the two moves below.
- The height coordinate, plant's default, does not move.
- **FF16 on the birth-date coordinate moves**, because part 8 changes how leaf area enters the light field. The one-species anchor goes from 17.14 to 18.01, and the two-species test from 12.05 and 16.59 to 12.07 and 16.76. At the default schedule FF16's error is now  $9\text{--}25\times$  larger. It falls at the square law, so a half-introduction run estimates it. Each move was registered before the references were blessed again.
- TF24's model version goes to v13. Its numbers change with the reverse pass, a storage pool offset, and a newborn's height found to roundoff.
- Under plant's default `Control()`, TF24's  $J$  on long drought is 0.45 on `develop`, against 12.75 converged. On this work's code the same defaults give 0.18, so the default settings, more than the model, carry that gap.

## Known gaps

- **Episodic rain needs finer introductions** for part 9's estimate to hold. Its error falls at the square law only once the spacing is below its dry spells (median 31 days), past 429 introductions.
- **Under constant rain the gradients move by up to  $3\epsilon$**  with where the later introductions sit; only the first year is graded finer.
- **A gradient blow-up with the soil stepped with the plants at a loose tolerance.** At tolerance  $1e-3$ , with the soil's tolerance factor at 10 and no inner steps, the resident's elasticities come out near  $10^{50}$  while  $J$  is right, and nothing refuses. Part 6, now the preset's default, removes it at every tolerance tried. Not yet root-caused.
- **How far one resident's recorded steps can serve** a moved resident or a distant invader is unmeasured; `diagnose_scm()` reports the distance without a bound.
- The branches sit on the fork's `develop` with `odelia 0.5.0`, and need a rebase onto upstream `develop` (`odelia 0.6.2`) before PRs open there.

## How to review: four groups, merged in this order

| group                                  | branches                                                        | read first                                                          | its test                                                                |
|----------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------|
| Parts 1–3:<br>counts and<br>gradients  | PLANT-93,<br>offspring-adjoint,<br>PLANT-95–99                  | stand_gradient,<br>run_mutant                                       | the reverse pass against<br>central differences; $J' = J$<br>to the bit |
| Part 4: the cut                        | state-weights,<br>PLANT-100–103,<br>odelia#53–54, phylloptim#17 | where a step is cut, and<br>how the reverse pass<br>crosses the cut | the 5% tolerance change;<br>the pass against<br>differences             |
| Parts 5–7: step<br>settings            | PLANT-104–106, odelia#55                                        | control_tf24,<br>control_window                                     | replays bit for bit; nothing<br>fails on five records                   |
| Parts 8–9:<br>nodes and<br>diagnostics | PLANT-107–110                                                   | the interpolated crowns;<br>diagnose_scm                            | the square law; the FF16<br>re-bless                                    |

Each group is bit-identical with its new controls off. Part 4 is the largest change, and changes what odelia hands plant after a step. Part 8 holds the one decision for maintainers: whether the interpolated crowns should apply to FF16's birth-date coordinate.